

Psychology in 2125: A Developmental Science of Living Sense-Making

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Abstract

A century from now, the computational framework at the center of psychology will no longer organize the discipline. As used here, "computational" names a broad information-processing ontology: meaning is specified in advance, rather than generated through the activity, history, and situation of living beings. This assumption fits engineered systems, but it misdescribes living ones. This article argues that three converging shifts are destabilizing that framework. The first undermines the computational account of inheritance: organisms develop through activity within configurations of constraints reconstituted across generations, not through the readout of genetic programs. The second displaces its account of psychological structure: psychological phenomena unfold as trajectories of coordination rather than as stable traits, with reliability and variability arising from the same generative process. The third sharpens the first two: artificial intelligence systems produce fluent performance without embodiment, developmental history, or any stake in the world. This contrast reveals what living organization contributes. The psychology implied by these shifts treats development as foundational, not as a subfield. It understands psychological constructs as patterns of activity that persons sustain over time, and is better positioned to address a century of changing human ecologies. This future psychology will be a developmental science of living sense-making.

Psychology in 2125: A Developmental Science of Living Sense-Making

A century from now, psychology will not rest on the foundations in place today. This article responds to the special issue's invitation not by predicting which specific ideas will prevail, but by identifying the structural conditions under which the currently dominant framework gives way. The computational framework that organizes contemporary psychology rests on assumptions that are under increasing empirical and conceptual pressure, and the forces driving its reorganization are already operating, whether or not the discipline has registered them. What follows traces what these pressures contest, what they make available, and what kind of psychology they imply.

Referring to the current framework as “computational” is meant to designate a broad information-processing ontology, not the use of formal or mathematical tools. Its core commitment is that meaning is specified in advance, rather than generated through the activity, history, and situation of living beings. This ontology has shaped theorizing and empirical work across much of the discipline. In the study of cognition, it appeared as the framing of thought as the processing of information, the content of which is settled in advance. The same architecture has been extended to social and emotional life, where cues and stimuli are treated as already meaningful inputs to be decoded or appraised by internal mechanisms. In developmental research, related assumptions appear in models that treat genes and environments as separable sources of information whose significance is given independently of the individual's activity. These are not separate commitments but facets of a single ontology. On this view, signals with fixed meanings are already present in the world, available to be received; genes or other purported codes carry preassigned significance; and internal mechanisms operate on inputs whose identity is defined independently of the organism's history.

Although it has been productive, the computational picture is increasingly difficult to sustain as an account of psychological life. In the laboratory, what a response will be taken to mean is fixed in advance by the experiment, not by what the organism's own activity makes matter. A similar move runs through developmental explanation. We begin with a robust outcome and then reconstruct the path that leads toward it. Reconstructing a path is not itself the problem; the trouble begins when the endpoint, once named, is treated as though it had been present from the start, directing the process toward itself. Development is then made to look like the delivery of something already specified, not the generative, historically situated activity of a living system. The problem, then, is not confined to one theoretical vocabulary. It is the same problem in different forms: the meaning of behavior and the form of a developmental outcome are both treated as if they preceded the living processes through which they are generated. The analysis that follows centers on development, because developmental explanation makes especially clear what is obscured when meanings, forms, and psychological phenomena are treated as prior to the processes that generate them.

Three empirical and conceptual shifts now contest the computational commitment and illustrate the shape of the alternative. First, evolutionary and developmental biology have challenged the view of development as the execution of a genetic program. Developmental systems theory and the Extended Evolutionary Synthesis have recast inheritance and development as the patterned activity of organisms, not the readout of preformed instructions. Second, idiographic and within-person methods have revealed that psychological phenomena unfold as trajectories of coordination rather than as stable traits that exist outside the activity that sustains them. Third, advances in artificial intelligence (AI) have demonstrated that fluent behavioral performance can be generated without a living being to produce it, exposing the

assumption that the surface form of behavior carries its meaning in the performance itself. The last shift plays a different role from the first two: rather than setting a further boundary, it sharpens the distinction between performance that can be simulated and sense-making that belongs to living systems.

Conceptual resources for moving beyond the computational framework exist, although they have been articulated in partially separate literatures. Developmental systems theory locates causation in the configurations of constraints through which organism–environment systems are reconstructed across generations, dissolving any partition between genetic and environmental contributions (Lickliter & Moore, 2023; Oyama et al., 2001). Process-relational developmental science highlights the inseparability of person and context and emphasizes circular over linear causality (Overton & Lerner, 2014; Overton, 2015; Witherington, 2011). Developmental research within the sociocultural tradition has shown that human cognition is not formed within the individual before meeting culture but is instead constituted through participation in shared practices (Cole, 1996; Rogoff, 2003; Vygotsky, 1978). Enactive approaches frame cognition as sense-making: the activity through which a living being, acting out of its own conditions of viability, brings forth a world that matters to it (Thompson, 2007; Varela et al., 1991).

These perspectives differ in emphasis and explanatory ambition, but each contests the assumption that psychological life can be specified apart from the historical, relational, and embodied processes through which it is sustained. They do not resolve into a tidier replacement for the currently dominant ontology, but they provide resources for an account of mind and development that is more biologically coherent and more exacting about what counts as explanation. Mind is not a system processing information about a world it stands apart from;

psychological life is the sense-making activity of living beings within conditions of life that are historically contingent and continually reconstructed.

Basis of the Forecast

This forecast rests on theoretical extrapolation from established empirical and conceptual work. Findings and frameworks from developmental biology and developmental systems theory, from idiographic methods, and from the distinction between living and artificial systems are treated as convergent evidence that narrows the range of coherent ontological positions the discipline can adopt. Two of these set boundary conditions on psychology's viable futures, identifying empirical constraints that any coherent successor framework must respect; the third, the contrast between living and artificial systems, serves as a diagnostic that sharpens what those conditions require.

The forecast is further grounded in Jablonka's (2023) extension of developmental systems theory and Waddington's epigenetic landscape metaphor to sociocultural systems, including scientific thought styles. On this account, theoretical persistence and change can be understood as products of self-sustaining networks of assumptions, practices, methods, training, and institutional arrangements. The century-scale horizon suits the analysis because the argument is not about the specific path psychology will take but about the conditions under which a disciplinary attractor becomes unstable. The forces identified here are already operating, and a century is sufficient time for their effects to alter the landscape of psychological science.

Rethinking Development

The first boundary condition concerns the causal vocabulary of development. For much of the twentieth century, both biology and psychology saw development as the sum of separate internal and external influences, with genes specifying form and the environment supplying

modification. Developmental psychology adopted this framing, treating innate tendencies and environmental inputs as separable contributions. This approach came to dominate the field, even though it was not the only available position. Piaget (1971) had argued that the structures of cognition are actively constructed through the organism's continuous interaction with its environment, although mainstream developmental psychology moved away from even this moderate commitment.

The assumption of separability is now being contested with new force, as developments in evolutionary and developmental biology give older critiques of genetic determinism and additive causation new empirical traction (Gottlieb, 2007; Lickliter & Moore, 2023). Taken together, the work reviewed below supports a different characterization of development: not as the execution of a genetic program or as an additive combination of separate internal and external causes, but as the activity of a developing organism within a structured field of constraints operating at multiple scales. The term “constraints” is used here to mean not external restrictions or determining causes, but enabling and limiting conditions that shape a field of possible activity, making some forms of organization possible, probable, or stable, while making others less likely (Juarrero, 2023).

Developmental biology provides a foundation for this shift. Morphogenesis involves the coupled activity of gene regulatory networks, tissue mechanics, chemical gradients, cellular architecture, and bioelectrical signaling, rather than proceeding through genetic specification alone (Forgacs & Newman, 2005; Newman & Bhat, 2009; Whited & Levin, 2019). Nor is this activity reducible to physical self-organization alone: cells sense, move, adhere, signal, and modify their surroundings, so that morphogenesis depends on agent-like cellular activity as well as on material and biochemical constraints. Gene expression itself is context-sensitive and

relational, shaped by cellular, physiological, epigenetic, and broader developmental conditions (Jablonka & Noble, 2019; E. F. Keller, 2011). On this view, DNA is not an instruction manual for development but one developmental resource among others, available only within the organized activity and conditions through which living form is produced (Nijhout, 1990; Noble, 2015; Oyama, 2000).

The reframing of DNA as a developmental resource sits within a broader transformation in the life sciences. The Modern Synthesis, with its emphasis on gene-centered accounts of evolutionary change, continues to organize much of evolutionary thought. But across multiple fronts, including developmental systems theory, ecological developmental biology, developmental plasticity research, and the Extended Evolutionary Synthesis, biologists and philosophers of biology increasingly treat gene-centered accounts as incomplete, emphasizing the developmental and ecological processes that such models leave out (Jablonka & Lamb, 2014; Laland et al., 2015; West-Eberhard, 2003). Psychology often inherited its assumptions about development and evolution from twentieth-century evolutionary theory, without tracking how contested those assumptions have since become.

Developmental systems theory provides a broader vocabulary for these insights, situating them within an integrated account of inheritance and development. Organisms inherit not only DNA but a suite of developmental resources, from cellular organization and ecological conditions to, in some lineages, social arrangements and cultural practices. What persists across generations is not a set of instructions but the life cycle itself, reconstituted each time from reliably available resources (Oyama et al., 2001). A stable developmental outcome can therefore recur without being specified in advance. It recurs because living systems actively maintain and rebuild forms of organization from one moment to the next and from one generation to the next.

A reliable outcome can be described once it has stabilized, but that description is not the process that produced it. A developmental account explains how forms of organization come to hold, not by assuming they were already encoded, but by tracing the activity through which they are assembled, sustained, and sometimes remade. That activity is never confined to a single phase of life or explanatory frame: cellular, bodily, ecological, and social constraints bear on development simultaneously, but their significance depends on how the organism's activity brings them into relation. In acting, the organism alters bodily dynamics, reorganizes local ecological arrangements, and reshapes patterns of interaction, so that each emerging form of coordination becomes part of the constraint structure that shifts the space of possibility for subsequent development, including what can be reconstructed across generations (Gilbert & Epel, 2015; Lewontin, 2000; Oyama et al., 2001).

In the human case, what is reconstructed is not a biological substrate to which culture is then added: the two have no prior existence apart to be combined. The slow maturation of the human cortex presupposes the cooperative caregiving arrangements that sustain a long childhood, and the human gut has been shaped by the food-preparation practices that altered its evolutionary trajectory. Cultural practices, social routines, and institutional arrangements are therefore developmental resources in the same sense as cellular organization and ecological conditions: they are reliably present across generations, they constrain what is possible without determining what occurs, and the child's growing participation in them reshapes the very constraints that shaped that participation (Cole, 1996; Rogoff, 2003; Vygotsky, 1978).

Psychological Phenomena as Trajectories

A second boundary condition follows from this reframing of development as a central form of explanation rather than a specialized domain within psychology: psychological

phenomena unfold as trajectories rather than as stable attributes that the individual possesses apart from the activity that sustains them. Yet for decades, psychological constructs have been treated as enduring properties, whether framed as traits, mechanisms, or latent structures. These assumptions were built into the field's dominant methods, organized around cross-sectional designs, group comparisons, and variable-centered modeling, which obscure the temporal organization of psychological life. The same habit splits what is in fact a single process into separate sources: species-typical reliability is explained through evolutionary or genetic programs and variability through individual differences, when reliability and variability instead arise together, through histories of activity within changing configurations of constraints. Cross-cultural developmental research makes this double aspect empirically visible (Henrich et al., 2010; H. Keller, 2007). Human experience, behavior, and cognition are sustained and reorganized through activity, not possessed as fixed properties apart from it.

If psychological phenomena are sustained through activity rather than possessed as stable attributes, then methods that compare fixed quantities across persons will misrepresent them. Within-person processes often differ fundamentally from between-person associations, and group-level regularities cannot be assumed to describe any actual individual (Molenaar, 2004; Nesselrode & Molenaar, 2010). Ergodic assumptions, long taken for granted in psychological modeling, do not hold for developing organisms whose trajectories depend on changing constraints, histories of interaction, and variable responses to perturbations. Recent work in longitudinal methodology makes this problem concrete: different statistical representations of change applied to the same data can produce contradictory inferences, revealing that the field's modeling choices embed unexamined assumptions about what developmental change is (McCormick & Bauer, 2024).

Intensive longitudinal and person-specific approaches show how patterns of activity develop and stabilize. As an example, Ram and Gerstorf (2009) developed a framework for treating within-person variability as developmentally informative rather than as noise. They showed how fluctuations on short timescales (minutes, days, weeks) carry information about individuals' capacities for change and about the regulatory processes through which those capacities are exercised. In turn, these short-timescale dynamics are coupled with long-term developmental change rather than separate from it. Periods of instability often precede transitions, while moments of stability reflect temporary alignments between the individual and the constraints shaping their activity.

A related finding from research on motor development illustrates this trajectory-based view, building on work that sees skills as emerging from the coupling of body, task, and environment (Thelen & Smith, 1994). The onset of walking is often treated as the maturational unfolding of an internal timetable, yet walking emerges through changing relations among bodily capacities, surfaces, opportunities for movement, and caregiver practices. Once walking stabilizes, it reorganizes the child's relation to the environment: objects, distances, people, and risks become available in new ways. The same posture-dependence shows in what fails to carry over: infants who have learned to avoid a drop-off while crawling do not simply transfer that avoidance when they begin to walk (Adolph & Hoch, 2019). An information-processing interpretation would expect transfer, since the infant has acquired knowledge about drop-offs. A constraint-based account instead makes sense of the lack of transfer: what was organized was not a stored representation of danger but a mode of coordination between perception and action within a specific postural configuration. On a new postural base, that perception-action relation must be assembled again.

This boundary condition undermines the idea that psychological constructs can be understood as stable traits or mechanisms (Richters, 2021). A trait-like measure captures only a statistical summary, often averaging over heterogeneous processes, situational contingencies, and developmental dynamics. Such measures may aid prediction under restricted circumstances, but they do not reveal the organization through which psychological life unfolds; they treat abstraction as if it were explanation. A trajectory perspective instead highlights how persons sustain coherence over time through bodily, behavioral, interpersonal, and cultural activity. Some patterns become more stable than others; when constraints shift or new possibilities arise, persons reorganize their activity.

Living and Artificial Systems

The two boundary conditions considered so far clarify what a science of living organization must explain. The rise of artificial intelligence (AI) systems capable of reproducing the surface forms of human behavior, without possessing the underlying organization that generated them, sharpens this clarification. The example of AI is not the deepest source of the reorientation proposed here; the deeper sources lie in development, living organization, and trajectories of activity. AI instead functions as a diagnostic pressure that makes precise what the two preceding boundary conditions require.

The distinction at issue is not between humans and a particular technology but between systems whose activity is grounded in their own conditions of viability and systems whose activity is not. Enactive cognitive science has theorized it in general terms, framing living systems as self-producing and self-maintaining, their activity answerable to their own continued existence rather than to externally imposed design objectives (Di Paolo, 2005; Maturana & Varela, 1980; Thompson, 2007; Varela et al., 1991). On this framing, the distinction does not

depend on how a specific engineered system is built or on how capable it becomes. Current large language models (LLMs) are a case in point: they reproduce fluent surface forms of language without the organization that supports reasoning, world knowledge, and functional language use in humans (Mahowald et al., 2024). For a discipline that has long treated behavioral and linguistic performance as its primary evidence for cognitive structure, this dissociation forces a choice. If fluent, contextually appropriate, apparently meaningful output can be generated without embodiment, developmental history, or any stake in the world, then either such outputs are sufficient evidence for mind, or psychology must distinguish behavioral form from sense-making as the activity of a living being grounded in its own stake in continuing to exist.

The present case of LLMs shows where the resemblance comes from. The training corpora on which these models have been built are not neutral data: they are the accumulated external traces of embodied, culturally situated, developmentally shaped human activity. The statistical regularities that these systems learn to reproduce are the shadows cast by living organization onto the surface of language. The system captures the shadows, but the substance remains outside its reach, because meaning arises in what living beings do as they act within the histories that shape them. Even as artificial systems change in ways that cannot be foreseen, the structural distinction holds. Engineered systems are not the kind of entities whose activity is grounded in their own continued existence, and they remain categorically different from living beings, regardless of their behavioral sophistication.

A field that models cognition on engineered performance rather than on the living organization that gives behavior its meaning will continue to lose its grip on its own object of study. The pressure toward this trajectory will grow as AI systems become more capable and more deeply woven into the environments in which human development unfolds. Meeting this

issue will be a major disciplinary challenge of the coming century: psychology must make the distinction between living sense-making and engineered performance central to its explanatory program, rather than treating it as a philosophical aside.

A Psychology Grounded in Living Organization: Implications for the Century Ahead

The two boundary conditions, together with the diagnostic contrast between living and artificial systems, converge on a shared conclusion: psychological life arises through the organization of a living system acting in a temporally extended world. Each challenges the computational ontology in a different register: that the form of a developmental outcome, the stability of a psychological trait, or the meaning of behavior could be specified without regard to the activity that generates it. They constrain the form a scientifically coherent discipline can take, and they clarify that psychological explanation must begin from the embodied, developmental, and relational processes through which activity becomes meaningful for living beings. This places meaning in the activity of living beings rather than in anything given in advance, and a psychology adequate to that claim needs to explain how such activity constitutes meaning.

The persistence of psychology's computational framework is not simply a matter of intellectual inertia. Dominant theoretical commitments persist through alignments among assumptions, methods, training programs, and empirical questions, producing a robust attractor. Jablonka (2023) applies a developmental systems analysis to the gene-centered Modern Synthesis and its resistance to the Extended Evolutionary Synthesis, drawing on Waddington's epigenetic landscape metaphor to characterize the dominant framework as a canalized attractor sustained by mutually reinforcing assumptions, methods, and institutional incentives. The computational framework in psychology presents a structurally parallel case: its attractor is robust because theoretical challenges can be incorporated without altering the assumptions,

methods, and institutional arrangements that sustain it. This is why earlier challenges to the computational framework did not fundamentally reorganize psychology. The resources reviewed at the outset of this article could be deflected or absorbed in weakened form: embodiment became an influence on cognition, context became a moderator of behavior, sociocultural participation became an external setting for development, and development itself became a background condition.

Jablonka's (2023) analysis frames shifts in attractors as long-running processes, beginning when the supports that sustain a framework start to change and unfolding over decades. The premise of the current paper is that the present pressures differ from earlier challenges because they are altering those supports. Shifts in developmental and evolutionary biology are eroding the biological credibility of instruction-based accounts of development and inheritance, and with it the broader assumption that living organization can be explained by pre-specified information. The emergence of practical idiographic methods is weakening the methodological reliance on pre-given traits, mechanisms, and group-level abstractions by making trajectories of activity increasingly tractable. The maturation of AI systems that generate fluent performance without living organization creates an existential disciplinary pressure that no amount of theoretical accommodation can resolve. Together, these forces are not simply arguments for change; they alter the conditions under which arguments operate. Over a century-scale horizon, alongside generational turnover, these are the kinds of structural forces under which a canalized attractor can shift.

A psychology reshaped by these pressures would look quite different from its current form. Rather than beginning from a taxonomy of cognitive mechanisms and the experimental designs that isolate them, the field would characterize developmental trajectories and ask how

patterns of coordination form, stabilize, and reorganize across the lifespan within particular cultural and ecological niches. Students entering the field would be trained in dense longitudinal and person-specific methods; their models would formalize constraint structures, not main effects; and their explanatory targets would be the conditions that sustain or disrupt human meaning-making. Memory, emotion, intelligence, and identity would be studied less as internal structures to be measured than as patterns of activity that persons sustain across time.

Neuroscience would be transformed correspondingly. The brain would be no less central, though what it means to study it would have changed. Rather than measuring neural responses to stimuli the experimenter has selected, neuroscientists would study ongoing neural activity as part of a developing organism acting in the world (Buzsáki, 2019). This would require methods suited to measuring such activity in freely moving individuals who are actively structuring their own environments and to studying how the brain's intrinsic dynamics are coupled with the dynamics of body and environment. The conventional partition between intrinsically driven and experience-driven aspects of brain development would give way to an account of neural organization as one strand of a developing organism–environment system (Marshall, 2024).

The commitments outlined here also shape what psychology can offer in a century of changing human ecologies. Climate disruption, mass migration, demographic aging, and digital technologies are altering the settings in which people sustain and remake their lives across generations. Such a discipline would be positioned to take this seriously in a way the current one is not. It would allow psychologists to study how lives remain coherent while being transformed by changing conditions, rather than treating stability and change as separate problems. Its trajectory-based methods would make actual lives, unfolding under altered ecological and social

arrangements, a tractable object of study. And its grounding in the distinction between living and engineered systems would keep psychological science answerable to its subject matter.

The future envisioned here is not one in which psychology adds development, embodiment, culture, or biology to its existing framework. It is one in which psychology redefines its object, and therefore the questions it can ask. The central question would no longer be where a capacity is located, which factor predicts an outcome, or whether an engineered performance matches a human one. The field would ask how ways of perceiving, acting, remembering, feeling, and relating become organized through development, and how they are sustained or remade as conditions change. Psychological phenomena are not information-processing operations inside individuals, stable attributes abstracted from groups, or surface performances detached from the processes that make them matter. They are patterns of activity through which persons with histories, dependencies, and stakes in what happens to them sustain meaning over time. A psychology adequate to the next century will be a developmental science of living sense-making.

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